

Functional genetic report · longevity

Kidney and Uric Acid

Sérgio F · Report date: 02-10-2026

Renal function (eGFR and chronic kidney disease), uromodulin, APOL1, urate transporters (SLC2A9, ABCG2, SLC22A12), gout, kidney stones and albuminuria. Only genes (chip + local imputation + polygenic scores).

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Summary

50

SNPs analyzed

40

in the chip

8

imputed ($DR2 \geq 0.8$)

2

SNP not found in the data

11

Polygenic scores

The kidney filters, regulates the balance of salts, acid and water and eliminates uric acid. Renal function falls with age in almost all people, and this decrease accelerates vascular and metabolic aging. This report combines the genes that weigh the most in renal function, uric acid, droplet and stones.

In synthesis: favourable renal genetics. The main eGFR score is at the 94th percentile (best predicted function), Sérgio F has 1 uromodulin protective haplotype, does not have APOL1 G1 nor the variant ABCG2 Q141K (the main genetic cause of high uric acid and gout). Genetic uric acid is typical (the 41st percentile) and the genetic risk of stones is medium to low. The weaknesses are small and dispersed (PRKAG2, A1CF, ALPL, HS6ST1).

Favourable

- **Renal function:** 253 eGFR score variants at the 94th percentile (301 variants in 80); IgA nephropathy at the 20th percentile.
- **Uromodulin:** 1 copy of the protective haplotype (UMOD rs12917707 GT, rs4293393 AG): less uromodulin, higher eGFR and less chronic kidney disease.
- **APOL1:** without G1 (rs73885319 AA, rs60910145 TT).
- **Uric acid and droplet:** ABCG2 Q141K and Q126X absent (rs2231142 GG); uric acid score at the 41st percentile and droplet at the 35 and 57 percentiles; HNF4G without risk.
- **Renal stones:** 42 loci score at the 17th percentile; SLC34A1 (rs6420094 AA, rs12654812 GG) and DGKH without risk alleles.

Unfavourable

- **PRKAG2 rs7805747 AA** (AMPK subunit): 2 copies of an uncommon risk allele of chronic kidney disease (28%).
- **Uric acid:** SLC2A9/GLUT9 heterozygous in the 3 SNPs (typical) and A1CF rs10821905 GA (infrequent risk allele).
- **Stones:** ALPL rs1256328 CT (infrequent risk allele); 5 599 urinary calculus score variants at the 79th percentile (at the limit of the typical).
- **Albuminuria:** HS6ST1 rs13427836 CT (small effect).
- **Not visible in the chip:** APOL1 G2 (deletion), CUBN A1690V and rare variants of monogenic kidney disease (polycystic kidneys, Alport, tubular nephropathy).

Because this matters to longevity: eGFR decreases ~ 1 mL/min/1.73 m² per year from 40 years and the kidney is one of the most aging organs. A high genetic base of eGFR is an advantage that protects itself mainly from aggressions (chronic NSAIDs, dehydration, sugar and high uric acid). Telmisartan and dapagliflozin, which already takes, are the drugs with more evidence of renal protection.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 40 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 1 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The IgA nephropathy score was reused from genomic imputation. «C_PEDRAS» is a simple score made with the leading variants by GWAS Catalog locus (less accurate than an PGS). In eGFR scores, a high percentile is favourable.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
eGFR (253 variants, Xie 2020)	PGS000303	247 of 253 (98%)	p94	Favourable
eGFR (301 variants, Monti 2024)	PGS004113	296 of 301 (98%)	p80	Typical / no judgement
Chronic kidney disease (241 variants)	PGS004224	237 of 241 (98%)	p61	Typical / no judgement
IgA nephropathy (55 variants)	PGS005284	53 of 55 (96%)	p20	Favourable
Uric acid (114 variants, Tin 2019)	PGS000126	110 of 114 (96%)	p41	Typical / no judgement
Drop (19 variants)	PGS003329	18 of 19 (95%)	p35	Typical / no judgement
Drop (G-PROB, 29 variants)	PGS000199	28 of 29 (97%)	p57	Typical / no judgement
Renal and ureteral stones (341 variants)	PGS001250	267 of 341 (78%)	p51	Typical / no judgement
Urinary calculus (5 599 variants)	PGS001864	5 415 of 5 599 (97%)	p79	Typical / no judgement
Renal stones (GWAS Catalog score, leader by locus)	C_PEDRAS	34 of 42 (81%)	p17	Favourable
Urinary albumin/creatinine (93 variants)	PGS000861	90 of 93 (97%)	p36	Typical / no judgement

Renal function (eGFR and chronic kidney disease)

Renal function measured by eGFR (estimated glomerular filtration rate) and risk of chronic kidney disease. Some genes alter only the markers (creatinine, cystatin C) and not the real function; they are marked as context.

Polygenic scores of this theme: eGFR (253 variants, Xie 2020) (PGS000303): the 94th percentile **Favourable**; eGFR (301 variants, Monti 2024) (PGS004113): the 80th percentile **Typical / no judgement**; Chronic kidney disease (241 variants) (PGS004224): the 61st percentile **Typical / no judgement**; IgA nephropathy (55 variants) (PGS005284): the 20th percentile **Favourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SHROOM3	rs17319721	GA	chip	A	1	0.41	SHROOM3 (podocytes): eGFR and chronic kidney disease (moderate evidence)	Typical / no judgement
PRKAG2	rs7805747	AA	chip	A	2	0.28	PRKAG2 (AMPK subunit): chronic kidney disease. (weak evidence)	Unfavourable
CERS2	rs267734	TT	chip	T	2	0.81	Ceramide synthase 2: eGFR. (weak evidence)	Typical / no judgement
DAB2	rs11959928	TA	chip	A	1	0.45	DAB2 (tubular endocytosis): eGFR. (weak evidence)	Typical / no judgement
SLC34A1	rs6420094	AA	chip	G	0	0.34	NaPi-IIa phosphate carrier: eGFR and calcium stones. (moderate evidence)	Favourable
MYH9	rs4821480	TT	imputation DR2=0.96	—	—	—	MYH9 (podocytes; LD with APOL1: kidney disease. (weak evidence)	Typical / no judgement
GATM	rs2467853	TG	chip	G	1	0.40	GATM (creatine synthesis): changes the creatinine produced, not renal function; it can bias the eGFR calculated by creatinine. (weak evidence)	Typical / no judgement
CST3	rs13038305	CC	imputation DR2=1.00	—	—	—	CST3 (cystatin C): alters cystatin C, not renal function; it can bias eGFR by cystatin. (weak evidence)	Typical / no judgement
SLC47A1	rs2289669	GA	chip	—	—	—	MATE1 (metformin tubular secretion and creatinine). (weak evidence)	Typical / no judgement

Favourable

- EGFR score (253 variants) at the 94th percentile; 301 score variants in the 80.
- IgA nephropathy at the 20th percentile.
- SLC34A1 rs6420094 AA (without risk allele).

Unfavourable

- PRKAG2 rs7805747 **AA** (2 copies of an uncommon risk allele).
- Chronic kidney disease score (241 variants) at the 61st percentile (typical).

Interpretation: Genetic basis favourable for filtration. The score of chronic kidney disease is typical because it mixes loci of eGFR with loci of other markers. GATM (creatine synthesis) and CST3 (cystatin C) touch the laboratory markers, not the function: it is useful to know this to interpret an eGFR by creatinine, especially with creatine supplements.

Uromodulin (UMOD)

Uromodulin (Tamm-Horsfall protein) is the most abundant protein in urine. The common variants of the promoter that increase it are associated with lower eGFR, more chronic kidney disease and more salt sensitivity; the minor haplotype (that low) is protective.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
UMOD	rs12917707	GT	chip	G	1	0.81	Uromodulin promoter: the risk allele increases uromodulin and is associated with lower eGFR and more chronic kidney disease. (moderate evidence)	Favourable
UMOD	rs4293393	AG	chip	A	1	0.81	UMOD: chronic kidney disease and eGFR. (moderate evidence)	Favourable
UMOD/PDILT	rs13333226	AG	chip	—	—	—	UMOD/PDILT: eGFR and kidney disease. (moderate evidence)	Typical / no judgement
UMOD	rs77924615	GG	chip	G	2	0.80	UMOD/PDILT: eGFR (leading variable in large GWAS). (moderate evidence)	Typical / no judgement

Favourable

- UMOD rs12917707 GT and rs4293393 AG: 1 copy of the protective haplotype (frequency ~20%).

Unfavourable

- UMOD rs77924615 GG and rs13333226 AG: common genotype (typical).

Interpretation: Sérgio F has 1 copy of the protective haplotype of UMOD, which in the literature is associated with higher eGFR and less chronic kidney disease (~10–20% less risk per copy). It is one of the explanations of the high eGFR score.

APOL1

The variants APOL1 G1 and G2 greatly increase the risk of kidney disease (risk ~7–10× with 2 risk alleles). They are rare in the reference population.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
APOL1 G1 (S342G)	rs73885319	AA	chip	G	0	0.00	APOL1 G1: with 2 risk alleles (G1/G2), ~7–10× risk of kidney disease. (strong evidence)	Favourable
APOL1 G1 (I384M)	rs60910145	TT	chip	G	0	0.00	APOL1 G1 (according to SNP haplotype). (strong evidence)	Favourable
APOL1 G2 (del N388/Y389)	rs71785313		SNP not found in the data	—	—	—	APOL1 G2: deletion of 6 pb (indel; the chips rarely read it). (strong evidence)	SNP not found in the data

Favourable

- APOL1 rs73885319 AA
- APOL1 rs60910145 TT

Unfavourable

- None.

Interpretation: Without G1 (2 SNPs genotyped, without the risk allele). G2 is a deletion of 6 pb that the chip does not read, but without G1 and with very rare G2 in the reference population, the risk APOL1 is very low.

Uric acid: transporters

Uric acid is the final product of purines. In excess, it crystallizes (dropper, stones) and activates the NLRP3 inflammasome; at normal levels it is an antioxidant. The kidney reabsorbs ~90% of the filtered urate (GLUT9/SLC2A9, URAT1/SLC22A12) and the intestine and kidney excreta it (ABCG2).

Polygenic scores of this theme: Uric acid (114 variants, Tin 2019) (PGS000126): the 41st percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SLC2A9 GLUT9	rs16890979	CT	imputation DR2=1.00	C	1	0.79	GLUT9: main urate resorption transporter; greater common effect on uric acid. (moderate evidence)	Typical / no judgement
SLC2A9 GLUT9	rs734553	GT	chip	T	1	0.76	GLUT9: uric acid and droplet. (moderate evidence)	Typical / no judgement
SLC2A9 GLUT9	rs13129697	TG	chip	T	1	0.72	GLUT9: uric acid. (moderate evidence)	Typical / no judgement
ABCG2 Q141K	rs2231142	GG	chip	T	0	0.09	ABCG2 Q141K: less intestinal and renal excretion of urate ~1.7x gout per copy; also more exposure to rosuvastatin. (strong evidence)	Favourable
ABCG2 Q126X	rs72552713	GG	chip	A	0	0.00	ABCG2 Q126X (null; rare): high uric acid and droplet. (strong evidence)	Favourable
SLC22A12 URAT1	rs505802	TC	chip	C	1	0.28	URAT1: resorption of urate in the proximal tubule. (moderate evidence)	Typical / no judgement
SLC22A12 URAT1	rs475688	CT	imputation DR2=1.00	—	—	—	URAT1: uric acid.	Typical / no judgement
SLC22A11 OAT4	rs17300741	AG	chip	A	1	0.46	OAT4: uric acid.	Typical / no judgement
SLC17A1/ A3 NPT1/ NPT4	rs1165205	TA	chip	—	—	—	NPT1/NPT4 carriers (urate excretion): uric acid and droplet. (moderate evidence)	Typical / no judgement
GCKR P446L	rs1260326	TC	chip	T	1	0.41	GCKR P446L: more hepatic glycolysis; triglycerides and higher uric acid, lower blood glucose. (moderate evidence)	Typical / no judgement
SLC16A9	rs12356193	AA	chip	A	2	0.83	MCT9: uric acid.	Typical / no judgement
PDZK1	rs12129861	CT	chip	C	1	0.54	PDZK1 (urate transporter gait): uric acid. (weak evidence)	Typical / no judgement
A1CF	rs10821905	GA	chip	A	1	0.20	A1CF: uric acid.	Unfavourable
MAF	rs7193778	TT	chip	C	0	0.14	MAF: uric acid.	Typical / no judgement
HNF4G	rs2941484	CC	chip	T	0	0.43	HNF4G: uric acid.	Favourable

Favourable

- ABCG2 Q141K absent (rs2231142 GG) and Q126X absent: normal urate excretion.
- Score of uric acid (114 variants) at the 41st percentile.
- HNF4G rs2941484 CC.

Unfavourable

- A1CF rs10821905 GA (infrequent risk allele).
- SLC2A9 heterozygous in the 3 SNPs (typical: half of the high urate alleles).

Interpretation: Typical genetic uric acid, without the major effect variant (ABCG2 Q141K). The heterozygous GLUT9 is the most common genotype. In practice, Sérgio F uric acid will depend more on the diet (fructose, purines), hydration and drugs (low-o dapagliflozin; bempedoic acid, if used, slightly above it).

Drop and inflammation by crystals

Gout: high uric acid is necessary, but not sufficient; the inflammatory response to crystals (inflamassoma NLRP3, IL-1β) decides the crisis.

Polygenic scores of this theme: Drop (19 variants) (PGS003329): the 35th percentile Typical / no judgement ; Drop (G-PROB, 29 variants) (PGS000199): the 57th percentile Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CNIH2	rs4073582	GA	chip	G	1	0.64	CNIH2: droplet.	Typical / no judgement
ALDH2	rs671	GG	chip	A	0	0.00	ALDH2*2: no relevance in those who do not drink alcohol. (moderate evidence)	Typical / no judgement
NLRP3 Q705K	rs35829419	CC	chip	A	0	0.05	NLRP3 Q705K: more active inflammasome (inflammatory response to urate crystals). (weak evidence)	Typical / no judgement
CARD8 C10X	rs2043211	AT	imputation DR2=0.99	T	1	0.33	CARD8 C10X: less brake of NLRP3 (hypothesis of more dripping inflammation). (weak evidence)	Typical / no judgement
IL1B	rs1143623	CG	chip	—	—	—	IL1B promoter (inflammatory response to crystals). (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: Droplet scores at the 35 and 57 percentiles (typical). NLRP3 Q705K absent; CARD8 C10X heterozygous (typical). ALDH2*2 absent (irrelevant without alcohol). No relevant genetic predisposition for gout.

Renal stones

Renal stones (mainly calcium oxalate): absorption and excretion of calcium (CLDN14, CASR, TRPV5), vitamin D (CYP24A1), phosphate (SLC34A1) and oxalate (AGXT, SLC26A6).

Polygenic scores of this theme: Renal and ureteral stones (341 variants) (PGS001250): the 51st percentile **Typical / no judgement**; Urinary calculations (5 599 variants) (PGS001864): the 79th percentile **Typical / no judgement**; Kidney stones (GWAS Catalog score, locus leaders) (C_PEDRAS): the 17th percentile **Favourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CLDN14	rs219780	CC	chip	C	2	0.79	Claudin-14: calcium resorption in the tubule; hypercalciuria and calcium stones. (moderate evidence)	Typical / no judgement
CASR A986S	rs1801725	GG	chip	—	—	—	Calcium receptor: serum calcium and stones. (weak evidence)	Typical / no judgement
CASR	rs7627468	GG	chip	A	0	0.24	Calcium receptor: stones. (weak evidence)	Typical / no judgement
CYP24A1	rs6127099	AT	chip	A	1	0.66	CYP24A1 (degrades active vitamin D): calcium stones; relevant with high doses of vitamin D. (moderate evidence)	Typical / no judgement
DGKH	rs1037271	TT	imputation DR2=1.00	C	0	0.40	DGKH: kidney stones.	Favourable
TRPV5	rs4236480	TC	imputation DR2=0.91	—	—	—	TRPV5 (distal calcium channel): stones. (weak evidence)	Typical / no judgement
SLC34A1	rs12654812	GG	chip	A	0	0.35	NaPi-IIa: stones and phosphate. (weak evidence)	Favourable
ALPL	rs1256328	CT	chip	T	1	0.18	Alkaline phosphatase: stones. (weak evidence)	Unfavourable
AQP1	rs1000597	TT	chip	—	—	—	Aquaporin 1: stones.	Typical / no judgement
SLC26A6	rs4973499	GT	imputation DR2=0.99	—	—	—	SLC26A6 oxalate carrier. (weak evidence)	Typical / no judgement
AGXT P11L	rs34116584	CT	chip	T	1	0.20	AGXT P11L (lower allele): less activity of AGT; more oxalate (clinical effect only in combination with other variants). (weak evidence)	Typical / no judgement

Favourable

- Rock score (42 loci) at the 17th percentile; PGS001250 in 51.
- SLC34A1 rs12654812 GG, DGKH rs1037271 TT, CASR rs7627468 GG.

Unfavourable

- ALPL rs1256328 CT (infrequent risk allele).
- PGS001864 (5 599 variants) at the 79th percentile (typical limit).
- CLDN14 rs219780 CC and CYP24A1 rs6127099 AT: common risk alleles (typical).

Interpretation: Genetic risk of medium to low stones. CYP24A1 is relevant with high-dose vitamin D (10 000 IU): rare variants of loss of function (which cause hypercalcemia with vitamin D) are not visible in the chip; the common variant rs6127099 is heterozygous (typical). AGXT P11L heterozygous effect only relevant in combination with other rare variants.

Albuminuria

Albuminuria: loss of albumin in urine, early marker of glomerular endothelial injury and cardiovascular risk.

Polygenic scores of this theme: Urinary albumin/creatinine (93 variants) (PGS000861): the 36th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CUBN A1690V	rs141640975		SNP not found in the data	A	—	—	Cubillin A1690V: benign albuminuria (less tubular albumin resorption); also B12. (moderate evidence)	SNP not found in the data
CUBN	rs1801239	TT	chip	C	0	0.09	Cubilin I2984V: albuminuria. (weak evidence)	Typical / no judgement
HS6ST1	rs13427836	CT	chip	T	1	0.14	HS6ST1 (heparan sulfate of basal membrane): albuminuria. (weak evidence)	Unfavourable

Favourable

• None.

Unfavourable

• HS6ST1 rs13427836 CT

Interpretation: Urinary albumin/creatinine score at the 36th percentile (typical). HS6ST1 CT with small effect. CUBN A1690V (benign albuminuria) not found in the data.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Vascular Aging	PRKAG2 AA; HS6ST1 CT	eGFR p94; UMOD protector (1 copy); APOL1 without G1	Favourable: the glomerulum is microvasculature.
Endoplasmic Reticulum Dysfunction / Protein Aggregation	—	Protective UMOD (less uromodulin poorly folded in the tubule)	Slightly favourable.
Lysosome dysfunction	—	APOL1 without G1 (APOL1 risk damages podocyte lysosomes)	Favourable.
Chronic Inflammation / Inflammation	SLC2A9 and A1CF (urate)	Normal ABCG2; NLRP3 Q705K absent; p35-57 gout	Typical uric acid: the high urate activates NLRP3.
Oxidative stress / Redox imbalance	—	Typical urate (antioxidant at the normal level)	Neutral.
Metabolic Homeostasis Decline / Insulin Resistance	GCKR TC (urate, triglycerides); PRKAG2 AA (AMPK)	—	Small weakness.
Nutrient Sensing Deregulation	PRKAG2 rs7805747 AA (AMPK γ 2 subunit)	—	It connects the kidney to AMPK; small effect.
Remodelling of the extracellular matrix	HS6ST1 CT	SHROOM3 and DAB2 typical	Neutral.
Endocrine dysregulation	CYP24A1 AT (typical)	CASR without risk	Neutral; attention to high vitamin D.
Peroxisomal dysfunction	AGXT P11L heterozygous (oxalate)	—	Small isolated effect.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Renal protection	Keep telmisartan and dapagliflozin, which already takes (both have strong evidence of renal protection). Avoid chronic NSAIDs (ibuprofen, naproxen, etc.): with telmisartan + dapagliflozin and CYP2C9 *1/*3, add renal risk. Paracetamol as the first line, as in the plan. suggestion (not applied)	high genetic eGFR to protect; Plan PGx
Hydration	2.5–3 L of water per day, more on hot and training days; dapagliflozin increases diuresis. Light urine is the target. suggestion (not applied)	Stones (ALPL, PGS p79); urate
Uric acid	Zero alcohol (already fulfills); little fructose free (coolers, juices, agave syrup); viscera and shellfish with moderation. Avoid very long fasts with ketosis and dehydration, which rise the uric acid. suggestion (not applied)	SLC2A9 heterozygous; A1CF GA
Vitamin C	Vitamin C of 500 mg of the plan (between meals) has a mild uricosuric effect. Do not rise to doses of several grams (sob urinary oxalate). suggestion (not applied)	Urate; AGXT P11L heterozygote
Vitamin D and calcium	Keep D3 10 000 IU as in the plan, with calcium by food and meals (alloys oxalate in the intestine and also reduces iron absorption). Avoid calcium in tablets outside meals. suggestion (not applied)	CYP24A1 (invisible rare variants); stones
Citrate	Lime or lime in water (natural citrate, inhibits stones). Potassium only by food, never in tablet (telmisartan), as in the plan. suggestion (not applied)	Stones
Salt and protein	Moderate salt and animal protein distributed throughout meals: both increase urinary calcium. suggestion (not applied)	CLDN14 CC; UMOD (salt sensitivity)
Creatine and eGFR	If taking creatine, creatinine rises without renal injury: to interpret renal function, prefer eGFR by cystatin C or combined. suggestion (not applied)	GATM and CST3 alter the markers
Lithium	Lithium 5 mg is a very low dose; maintaining the plan of pause in days of vomiting or diarrhea (telmisartan rises lithium). suggestion (not applied)	Coordination with the plan
What not to do	Do not use telomerase activators. Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed). suggestion (not applied)	V2 plan rules

Limitations

- APOL1 G2 is a deletion and is not read by the chip; the risk APOL1 is based on the absence of G1 and the rarity of G2 in the reference population.
- Monogenic kidney diseases (polycystic PKD1/PKD2, Alport syndrome, tubular nephropathy, primary hyperoxaluria) are not covered.
- EGFR scores estimate the mean genetic function, not that of the moment; the real function depends heavily on drugs, hydration and metabolism.
- «C_PEDRAS» is a simple score (leading variables of GWAS Catalog, published coefficients); it is less accurate than PGS.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use blood or urine analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Frequency of alleles calculated locally in the reference population (n=633).
- Effect alleles: cured from the literature or removed from the GWAS Catalog (local copy of 13-09-2026), with the verified effect.
- Scores: PGS000303 (Xie 2020), PGS004113 (Monti 2024), PGS004224 (Lan 2023), PGS005284 (IgA nephropathy), PGS000126 (Tin 2019), PGS003329 (Sumpter 2022), PGS000199 (Knevel 2020), PGS001250 (Tanigawa 2022), PGS001864 (Privé 2022), PGS000861 (Aly 2021) and a simple score of stones (42 loci from GWAS Catalog). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Köttgen 2009 and Wuttke 2019 (eGFR, Nat Genet); Trudu 2013 (UMOD, Nat Med); Genovese 2010 (APOL1, Science); Tin 2019 (urate, Nat Genet); Major 2024 (drip); Woodward 2009 (ABCG2); Thorleifsson 2009 (CLDN14); Howles 2019 (stones); Heerspink 2020 (DAPA-CKD, NEJM).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).